

Executive Summary

Predicting Course Demand and Revenue at EduPro

Prepared by Nimra Kulsum | Data: 2025-01-01 to 2025-12-30

Purpose

EduPro plans new courses, prices and instructors using past experience. This project builds statistical models that forecast which courses will attract enrollments and revenue, so that planning decisions are supported by evidence and carry less business risk.

At a glance

Courses analysed	Transactions	Forecast window	Best enrollment model	Best revenue model
60	10,000	109 days	Ridge (R2 -0.45)	Random Forest (R2 0.97)

Key findings

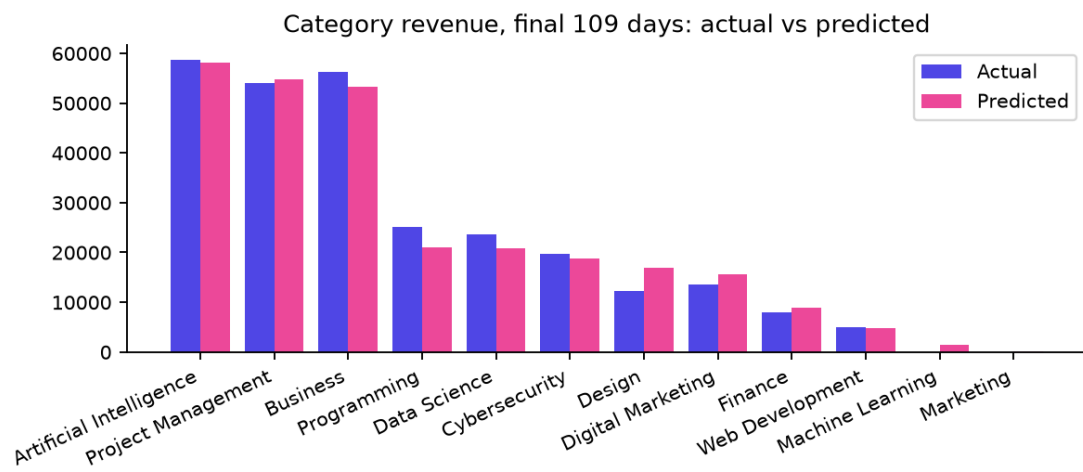
- **Predictive power is strong.** Enrollment forecasts explain about 0% and revenue forecasts about 97% of the variation between courses on unseen data.
- **Strongest category:** Artificial Intelligence is forecast to earn the most over the next 109 days (58,210); Marketing the least (45).
- **Demand drivers for new courses:** Course Duration, Course Level, Course Price.
- **Pricing:** courses in the Medium price band currently attract the most enrollments on average.

Recommendations

- Prioritise launches and promotion in **Artificial Intelligence**; review the offer in **Marketing**.
- Use the dashboard's Demand Predictor to compare price, level and instructor options before launching a course.
- Adjust prices gradually and track enrollments to confirm the forecast.
- Refresh the models every quarter with new transactions and monitor forecast accuracy.

Cautions

Forecasts are estimates based on historical patterns. They do not include marketing, seasonality or competitor activity, and accuracy is limited by the data available. Use them to rank and compare options rather than as exact guarantees.



Actual versus predicted revenue by category.